

RECIPE FINDER

MINI PROJECT REPORT

submitted by

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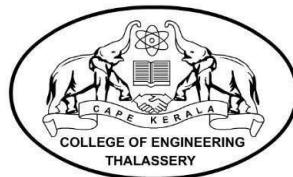
To

The APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the Degree
of

Bachelor of Technology

in

Information Technology



Department of Information Technology

College of Engineering Thalassery

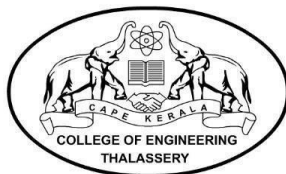
(Under Co-operative Academy of Professional Education)

Established by the Government of Kerala

Kannur Dt. - 670107

JULY 2024

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CERTIFICATE

This is to certify that the project preliminary work entitled

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to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Information Technology is a bonafide record of the project preliminary work carried out by them under our guidance and supervision.

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ACKNOWLEDGEMENT

We are greatly indebted to all those who have rendered help for successful completion of this project preliminary work.

First, we wish to extend our sincere gratitude to our Principal, **Prof. (Dr.) Rajeev P** for the help extended by her in providing all the college facilities for our project design.

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We are thankful and profoundly indebted to our Project Guide **Asst Prof Fahida A K**, Assistant Professors in Department of Information Technology for her exemplary guidance, criticism, monitoring and constant encouragement throughout the course and for showing us the right way.

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ABSTRACT

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The Food Recipe Finder software is designed to assist users in discovering meal recipes based on the ingredients they have on hand. In today's fast-paced world, minimizing food waste and making efficient use of available resources are more important than ever. This software aims to address these needs by providing a simple and intuitive platform where users can input the ingredients they currently have, and receive a curated list of recipes that can be prepared using those ingredients.

The core functionality of the Food Recipe Finder includes an extensive database of recipes, a user-friendly interface for ingredient input, and a sophisticated matching algorithm that suggests recipes based on the provided ingredients. This software not only helps users save time and effort in meal planning but also encourages creativity in the kitchen by offering diverse recipe options that users might not have considered otherwise.

By leveraging modern technologies and culinary databases, the Food Recipe Finder ensures that users can quickly and easily find meal ideas, reduce food wastage, and enjoy a more varied and satisfying diet. This project demonstrates the practical application of software development in solving everyday problems, enhancing user convenience, and promoting sustainable practices in food consumption.

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ABBREVIATIONS

- CSS Cascading Style Sheet
- HTML Hyper-Text Markup Language
- MYSQL Structured Query Language

CHAPTER 1

INTRODUCTION

The Food Recipe Finder software is designed to tackle the common problem of deciding what to cook based on the ingredients available at home. In many households, meal planning can be a stressful and time-consuming task, often resulting in food waste and limited culinary variety. By providing a platform where users can input their ingredients and receive tailored recipe suggestions, the Food Recipe Finder aims to simplify meal planning, promote sustainability, and enhance the cooking experience.

The software leverages an extensive recipe database, ensuring a wide array of meal options that cater to various dietary preferences and cultural cuisines. An advanced ingredient matching algorithm ensures that users receive accurate and relevant recipe suggestions, maximizing the use of their available ingredients and reducing the need for additional grocery shopping. The user-friendly interface makes it easy for users to input their ingredients and navigate through the recipe options, offering a seamless and enjoyable experience.

Furthermore, the Food Recipe Finder promotes sustainable cooking practices by helping users minimize food waste and make the most of their existing supplies. This project highlights the potential of technology to address everyday challenges, making cooking more efficient, creative, and environmentally friendly. Through the Food Recipe Finder, users can discover new recipes, reduce food waste, and enjoy a more varied and satisfying diet.

CHAPTER 2

SYSTEM STUDY

On studying the system, we came to know that the existing system has a lot of drawbacks;

1. Limited database of recipe
2. Time consuming
3. Inaccurate ingredient matching
4. Poor user interface

2.1 EXISTING SYSTEM

Existing systems for finding meal recipes based on available ingredients often fall short in several critical areas. Many have a limited recipe database, leading to repetitive and uninspiring meal suggestions that fail to cater to diverse dietary needs and preferences. The ingredient matching algorithms are frequently inaccurate, suggesting recipes that require additional ingredients not listed by the user, thus defeating the purpose of the system. Additionally, the user interfaces of these systems are often poorly designed, making it cumbersome for users to input ingredients and navigate through recipe options. Moreover, a significant lack of personalization results in irrelevant and generic recipe recommendations, reducing user engagement and satisfaction. Overall, these shortcomings hinder the practicality and appeal of existing recipe finder systems.

2.2 PROPOSED SYSTEM

The proposed Food Recipe Finder software will offer a range of advanced features designed to enhance the user experience and address the limitations of existing systems. It will include a comprehensive and diverse recipe database, ensuring a wide variety of meal options. An advanced ingredient matching algorithm will accurately pair user-inputted ingredients with suitable recipes, reducing the need for additional ingredients. The user-friendly interface will simplify ingredient input and recipe navigation, making the process intuitive and efficient. Additionally, the system will provide personalized recipe recommendations based on user preferences and dietary restrictions, ensuring relevant and engaging suggestions. Overall, the proposed system aims to simplify meal planning, reduce food waste, and promote sustainable cooking practices.

CHAPTER 3

SRS - SOFTWARE REQUIREMENTS SPECIFICATION

INTRODUCTION

This document serves as the Software Requirements Specifications (SRS) for the Recipe Finder website, aimed at providing users with a streamlined experience in discovering and preparing recipes. The website is designed to enable users to search for recipes based on available ingredients, dietary preferences, and cooking time, fostering efficiency and creativity in meal planning. By leveraging a comprehensive recipe database and intuitive user interface, the Recipe Finder website aims to empower users to explore diverse cuisines, reduce food waste, and enhance their culinary experiences. This document outlines the functional and non-functional requirements essential for the development and deployment of the Recipe Finder website, ensuring it meets the needs and expectations of its users effectively.

1.1 PURPOSE

The purpose of a recipe finder website is to simplify meal planning and enhance culinary exploration by offering users a streamlined platform to discover diverse recipes based on their preferences and available ingredients. By providing a user-friendly interface and a comprehensive database of recipes, these websites empower users to efficiently search for and explore new meal ideas, whether they are seeking specific dietary options, cultural cuisines, or simply looking to reduce food waste. Additionally, recipe finder websites often foster community engagement through user-generated content, encouraging sharing of recipes, tips, and feedback among cooking enthusiasts, thus creating a dynamic and supportive culinary community.

1.2 INTENDED AUDIENCE AND READING SUGGESTIONS

The intended audience of the Food Recipes Finder website includes cooking enthusiasts, home chefs, busy individuals seeking meal inspiration, and anyone interested in exploring diverse culinary options based on available ingredients.

1.3 PRODUCT SCOPE

This application is built such a way that it suits for all type of culinary ingredients .So every effort is taken to implement this project in this website, on successful implementation in this website, we can

target many pioneer food enthusiast. Main modules of the project: This project have the following modules:

Ingredient list

User details

Recipe collection details

Review

1.3 REFERENCE

- <https://github.com>
- <https://realfood.tesco.com/what-can-i-make-with.html>

2 Overall Description

2.1 PROJECT PRESPECTIVE

From the project perspective, our Recipe Finder aims to revolutionize how users discover and prepare meals. Centered on user needs, it prioritizes intuitive recipe search functionalities, personalized recommendations, and a robust recipe database. Technically, it emphasizes scalability to accommodate a growing user base and seamless integration with external APIs for ingredient data and nutritional information. Operationally, the focus is on efficient content management, user support, and continuous improvement based on user feedback. Regulatory compliance ensures adherence to data protection laws and copyright regulations, while marketing strategies aim to build user engagement through digital channels and partnerships. Ultimately, our Recipe Finder seeks to redefine culinary exploration, foster community interaction, and become a trusted resource for culinary enthusiasts.

2.2 PRODUCT-FUNCTION

Class of use cases	Use cases	Descriptions
Use cases related to registration of user	1. Register the user by himself.	1. store personal, contact details of the user.
Use cases related to system authorization of the user	1. Login of user. 2. Change Password of the user.	1. Login user into the system. 2. Change the Login Password of the user of the system.
Use cases related to displaying recipe	Send recommended food recipes to the user.	Display the recipe and add

2.3 USER CLASSES AND CHARACTERISTIC

The Recipe Finder website serves a diverse range of user classes, each with unique characteristics and preferences. Casual Cooks appreciate simple navigation and easy-to-follow recipes for everyday meals. Home Chefs seek diverse recipe options with detailed instructions and enjoy exploring new culinary techniques. Professional Cooks require advanced recipes and may contribute expertise to the community. By understanding and catering to these user classes, the Recipe Finder aims to provide a personalized and enriching culinary experience for all users.

2.4 OPERATING ENVIRONMENT

Particulars	Client System	Server System
Operating system	Windows/ Linux	Windows/Linux
Processor	Pentium ,intel....	Pentium ,intel
HDD		100gb

2.5 Design and implementation constraints

The design and implementation of the Recipe Finder website are constrained by several factors. Technologically, the choice of a suitable stack must align with existing infrastructure and accommodate future scalability needs. Performance requirements dictate efficient data retrieval and processing to ensure seamless user experience and system responsiveness under varying loads. User interface design constraints focus on achieving accessibility compliance, responsive design across devices, and intuitive navigation. Content management constraints encompass ensuring copyright compliance for recipe content and moderating user-generated contributions effectively.

2.6 User Documentation

For user documentation and information please consult section 4 external interface requirement and attached user manual.

2.7 Assumptions and Dependency

There are:

- Every user has a mobile phone.
- The details of ingredients and recipe
- The system database will be accessible in real time.

3 External Interface Requirement

The system is basically running on the official website of the govt, blood bank. Mainly there are 2 actors in the system. The system provides some advance features to the system admin than the donor. If the system admin logs in, the system interface provides some main command buttons to the admin.

- Change login password.
- Edit donor profile details.
- Search donors for a exact blood group and send messages.
- Print statements.
- Update the database .and Send blood testing details.
- Search details from the database .If the donors log in, the system will provide another different interface with different commands.
- Change login passwords.
- Edit personal .contact details
- Details related to contributions to donations .Future blood donation details.
- Withdraw name from the system.

3.1 User Interfaces

It has been required that every form's interface should be user friendly and simple to use.

3.2 Hardware Interfaces

- 1GHz or High processor
- 512 MB RAM
- 1 GB Hard Disk

3.3 Communication Interface

There are:

- Should have a proper internet connection.
- The response time for occurs a change will be more than 4 seconds.

- The response time for access the database will be no more 5 seconds

4 SYSTEM FEATURES

4.1. System Feature 1

1. Recipe Search and Filtering: Users can search for recipes based on ingredients, cuisine types, dietary preferences (e.g., vegetarian, gluten-free), meal categories (e.g., breakfast, dinner), and cooking time.
2. Recipe Details: Each recipe listing includes detailed instructions, ingredient quantities, nutritional information, cooking tips, and user reviews to aid in decision-making.
3. User Profiles: Users can create profiles to save favorite recipes.
4. Content Management: Backend tools for administrators to manage recipe database, user accounts, content moderation, and analytics to monitor user engagement and platform performance.
5. Accessibility and Security: Implementation of accessibility features for users with disabilities and robust security measures to protect user data and ensure platform integrity.
6. Support and Help Resources: Provision of user support through customer service channels to assist users with platform usage and troubleshooting.

4.2 Functional Requirements

- User Registration and Authentication
- Recipe Search and Filtering
- Recipe Details and Viewing
- User Interaction
- Shopping List and Meal Planning
- Admin and Content Management

5 OTHER NON-FUNCTIONAL REQUIREMENTS

There are the following non-functional requirements of Recipe Finder system.

5.1 Performance Requirements

1. Response Time: Ensure quick response times for recipe searches and page loads, with most operations completing within a few seconds.
2. Scalability: The system should handle increasing numbers of users and recipes without significant degradation in performance.
3. Concurrency: Support multiple users accessing and interacting with the website concurrently without slowdowns or errors.
4. Database Performance: Efficient database queries and indexing to retrieve recipe details and user data promptly.
5. Caching: Implement caching mechanisms to reduce database load and improve response times for frequently accessed data.

5.2 Safety Requirements

1. Data Encryption: Encrypt sensitive user data such as login credentials, personal information, and payment details to prevent unauthorized access.
2. Secure Authentication: Implement strong authentication mechanisms to verify user identities securely.

.5.3 Security Requirements

1. Access Control: Implement access control mechanisms to restrict unauthorized access to sensitive resources and functionalities within the platform.
2. User Privacy: Ensure user privacy by implementing privacy-enhancing features such as anonymization and pseudonymization of user data.

5.4 Software Quality Attributes

The system will pose some quality attributes to the users:

1. Reliability: The system has the ability to work all the times without failures apart from network failure. A donor can have the faith on the system. The authorities will keep the privacy of all donors in a proper manner. When the doctors found any disease in the testing stage after providing relevant details to the donor the system keeps the secretively of the donor.

2. Robustness: The entire system includes every function which is always help to the system to work correctly and strongly in all conditions.

3. Interoperability: The system includes the ability to work with the other applications which are also run on the same website.

4. Maintainability: The system shall provide the capability to back up the database. HMS will be maintainable as long as there are no Hardware & Software problems. Also Database should be updated.

CHAPTER 4 SYSTEM DESIGN

4.1 Gantt chart

TITLE	START DATE	DUE DATE	PREDECE SSORS	MARCH	APRIL	MAY	JUNE	JULY	
Project with dependencies									
Feasibility study	14/03/24	21/03/24		Feasibility study					
Required gathering	28/03/24	04/04/24			Requirement gathering				
SRS	25/04/	02/05/24				SRS preprati			

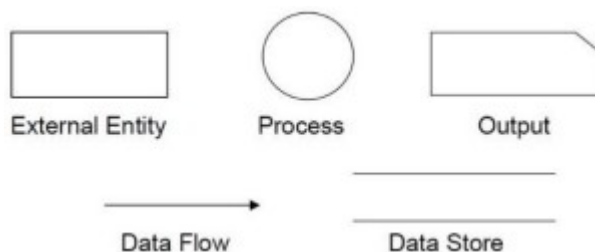
preparation	24					on			
Design	16/05/ 24	23/05/24					Desig n		
Report Prepration	10/07/ 24	10/07/24						Report	

4.2 DATA FLOW DIAGRAM (DFD)

System design is the process of designing the architecture, components and interfaces for a system so that it meets the end-user requirements. We have created Data Flow Diagram level 0 and 1, Use Case diagrams, Class diagram, Sequence diagram, Activity diagram, Data Flow diagram usually represent systems and processes that would be hard to describe in a chunk of text. While Use Case diagram is the written description of how users will perform tasks on your website. Class diagrams are fundamental to the object modeling process and model the static structure of a system. Sequence diagram is interactive based and Activity diagram a behavioral diagram.

The DFD (also known as a bubble chart) is a hierarchical graphical model of a system that shows the different processing activities or functions that the system performs and the data interchange among these functions. Each function is considered as a processing station (or process) that consumes some input data and produces some output data. The system is represented in terms of the input data to the system, various processing carried out on these data, and the output data generated by the system. A DFD model uses a very limited number of primitive symbols to represent the functions performed by a system and the data flow among these functions.

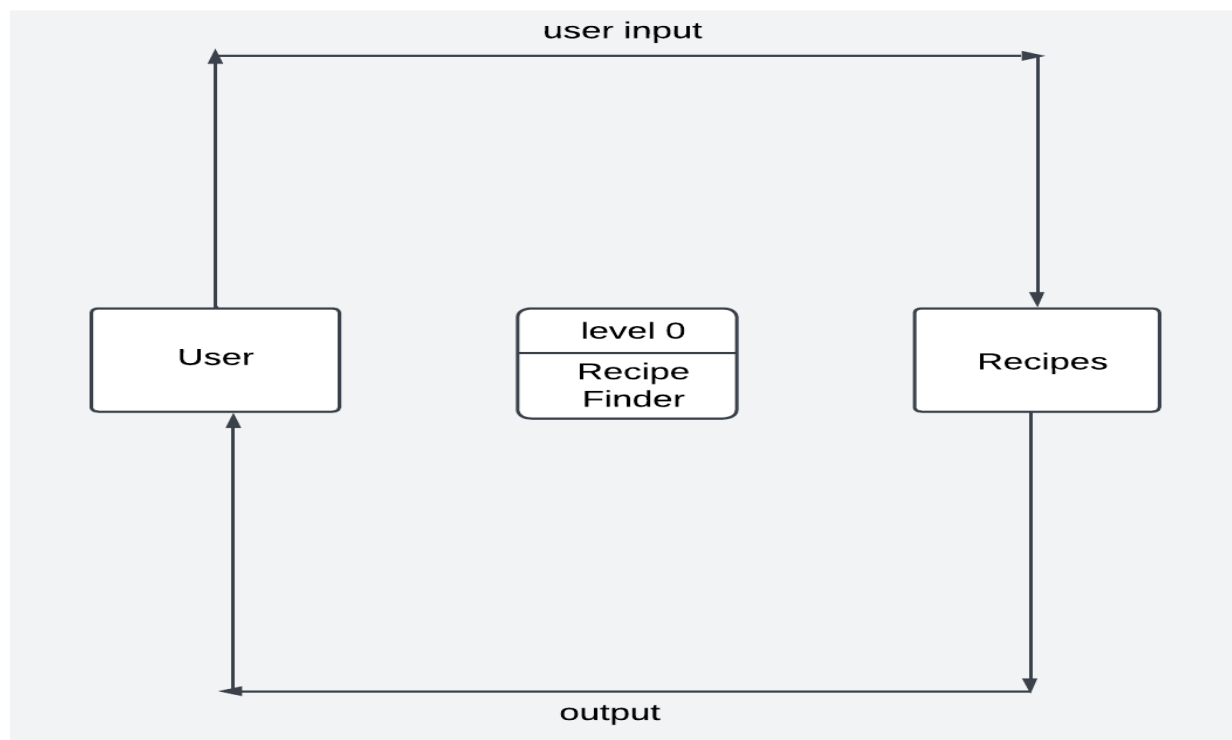
4.3 DFD Symbols



A data flow diagram can dive into progressively more detail by using levels and layers, zeroing in on a particular piece. DFD levels are numbered 0, 1 or 2, and occasionally go to even Level 3 or beyond. The necessary level of detail depends on the scope of what you are trying to accomplish.

Level 0 DFD

DFD Level 0 is also called a Context Diagram. It's a basic overview of the whole system or process being analyzed or modeled. It's designed to be an at-a-glance view, showing the system as a single high-level process, with its relationship to external entities. It should be easily understood by a wide audience, including stakeholders, business analysts, data analysts and developers. It's designed to be an at-a-glance view, showing the system as a single high-level process, with its relationship to external entities. In this level capturing images as input. We use a mobile camera for scanning images then it is processed by using image processing technique. The output, that is whether the currency is genuine or not in audio format



Level 1 DFD

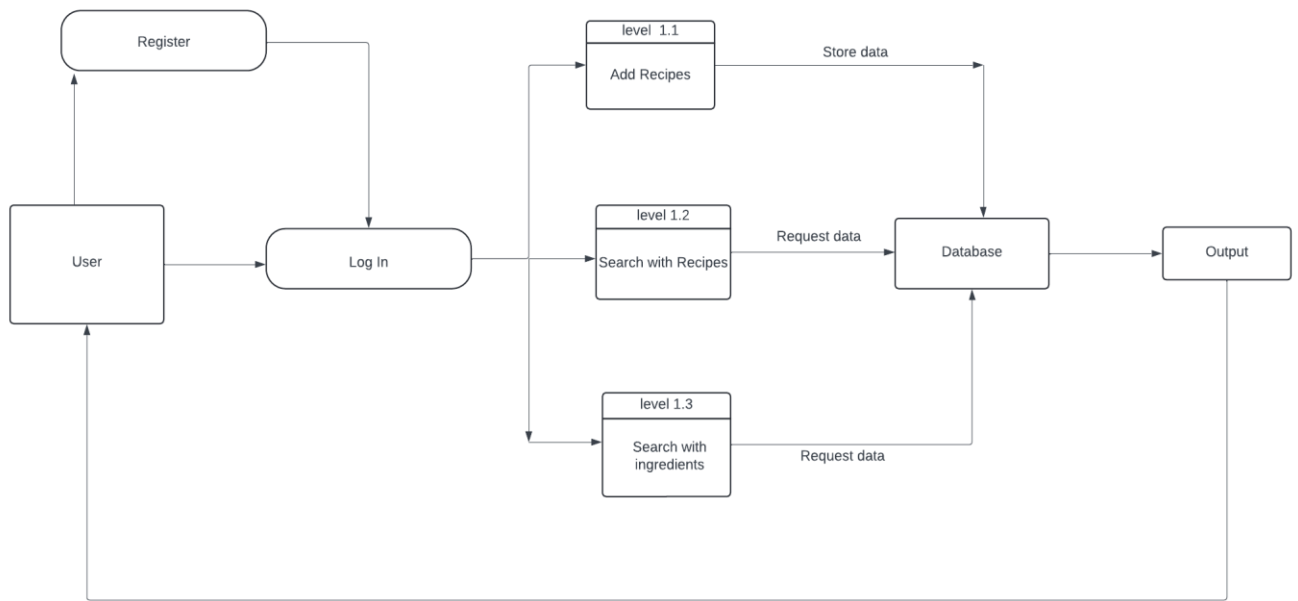
DFD Level 1 provides a more detailed breakout of pieces of the Context Level Diagram. You will highlight the main functions carried out by the system, as you break down the high-level process of the Context Diagram into its subprocesses.

In 1-level DFD, the context diagram is decomposed into multiple bubbles/processes. In this level, we highlight the main functions of the system and breakdown the high-level process of 0-Level DFD into subprocesses. In this level scanned images are processed by using image processing technique. The features of acquired image are extracted and stored in the database. From the extracted features, some relevant features are selected for classification. Features extracted from the training set and the extracted features from acquired image is then compared using classifiers then the result is presented to user in audio format.

DFD Level 2 then goes one step deeper into parts of Level 1. It may require more text to reach the necessary level of detail about the system's functioning.

We have to keep in mind following rules and tips:

- A context diagram should depict the system as a single bubble.
- All external entities interacting with the system should be represented only in the context diagram. The external entities should not appear at other levels of the DFD.
- It is a common oversight to have either too less or too many bubbles in a DFD. Only 3 to 7 bubbles per diagram should be allowed, i.e. each bubble should be decomposed to between 3 and 7 bubbles.
- Each process should have at least one input and an output.
- Each data store should have at least one data flow in and one data flow out.
- Data stored in a system must go through a process.
- All processes in a DFD go to another process or a data store.



5.1 USE CASE DIAGRAM

Use case diagrams are considered for high level requirement analysis of a system. When the requirements of a system are analyzed, the functionalities are captured in use cases.

We can say that use cases are nothing but the system functionalities written in an organized manner. The second thing which is relevant to use cases are the actors. Actors can be defined as something that interacts with the system.

Actors can be a human user, some internal applications, or may be some external applications. When we are planning to draw a use case diagram, we should have the following items identified.

- Functionalities to be represented as use case
- Actors
- Relationships among the use cases and actors

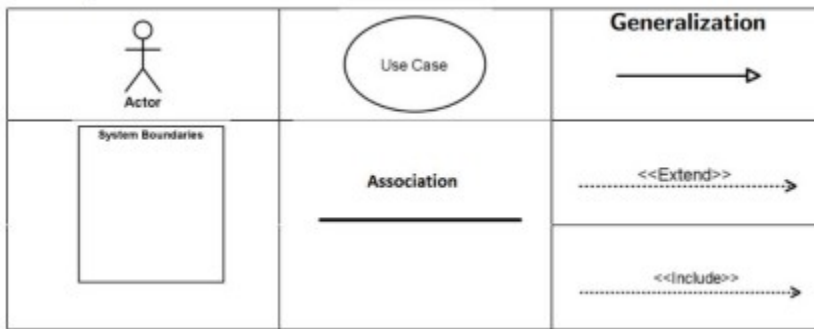
Use case diagrams specify the events of a system and their flows. But use case diagram never describes how they are implemented. Use case diagram can be imagined as a black box where only the input, output, and the function of the black box is known. These diagrams are used at a very high level of design. This high-level design is refined again and again to get a complete and practical picture of the system.

A use case is a sequence of transactions performed by a system that yields an outwardly visible, measurable result of value for a particular actor. A use case typically represents a major piece of functionality that is complete from beginning to end.

In UML, a use case is represented as an ellipse. Give your use case a unique name expressed in a few words (generally no more than five words). An actor represents whoever or whatever (person, machine, or other) interacts with the system. The actor is not part of the system itself and represents anyone or anything that must interact with the system to:

- Input information to the system;
- Receive information from the system; or
- Both input information to and receive information from the system.

Arrows and lines are draw between actors and use cases and between use cases to show their relationships. The default relationship between an actor and a use case is the «communicates» relationship, denoted by a line.

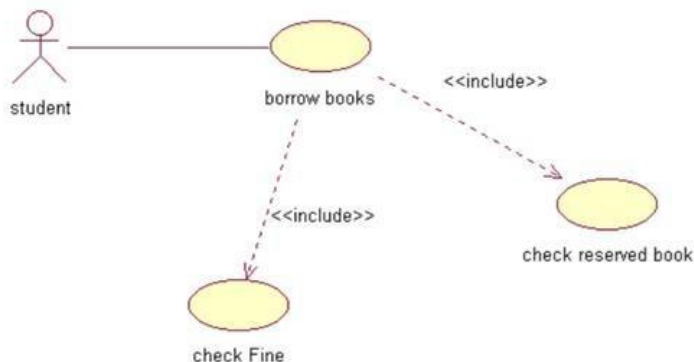


There are two other kinds of relationships between use cases (not between actors and use cases) that you might find useful. These are «include» and «extend». The «include» relationship is not the default relationship. Therefore, in a use case diagram, the arrow is labeled with «include» when one use case makes full use of another use case. You use the «extend» relationship when you are describing a variation on normal behavior or behavior that is only executed under certain, stated conditions.

It is common to be confused as to whether to use the include relationship or the extend relationship. Consider the following distinctions between the two:

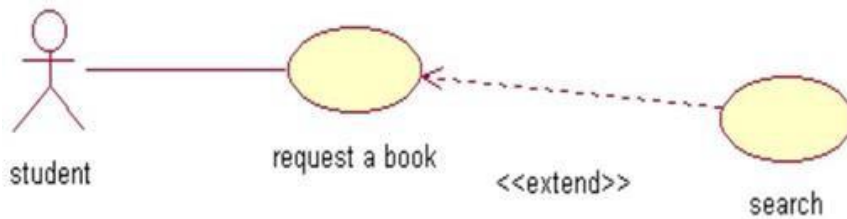
- Use Case X includes Use Case Y:

X has a multi-step subtask Y. In the course of doing X or a subtask of X, Y will always be completed.



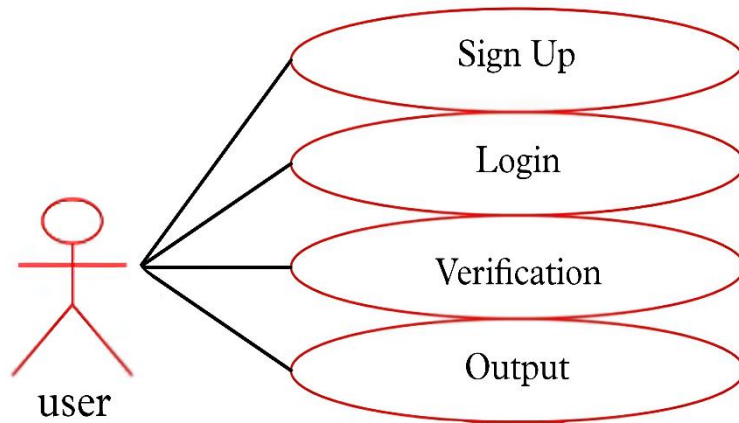
- Use Case X extends Use Case Y:

Y performs a sub-task and X is a similar but more specialized way of accomplishing that subtask (e.g. closing the door is a sub-task of Y; X provides a means for closing a blocked door with a few extra steps). X only happens in an exception situation. Y can complete without X ever happening.



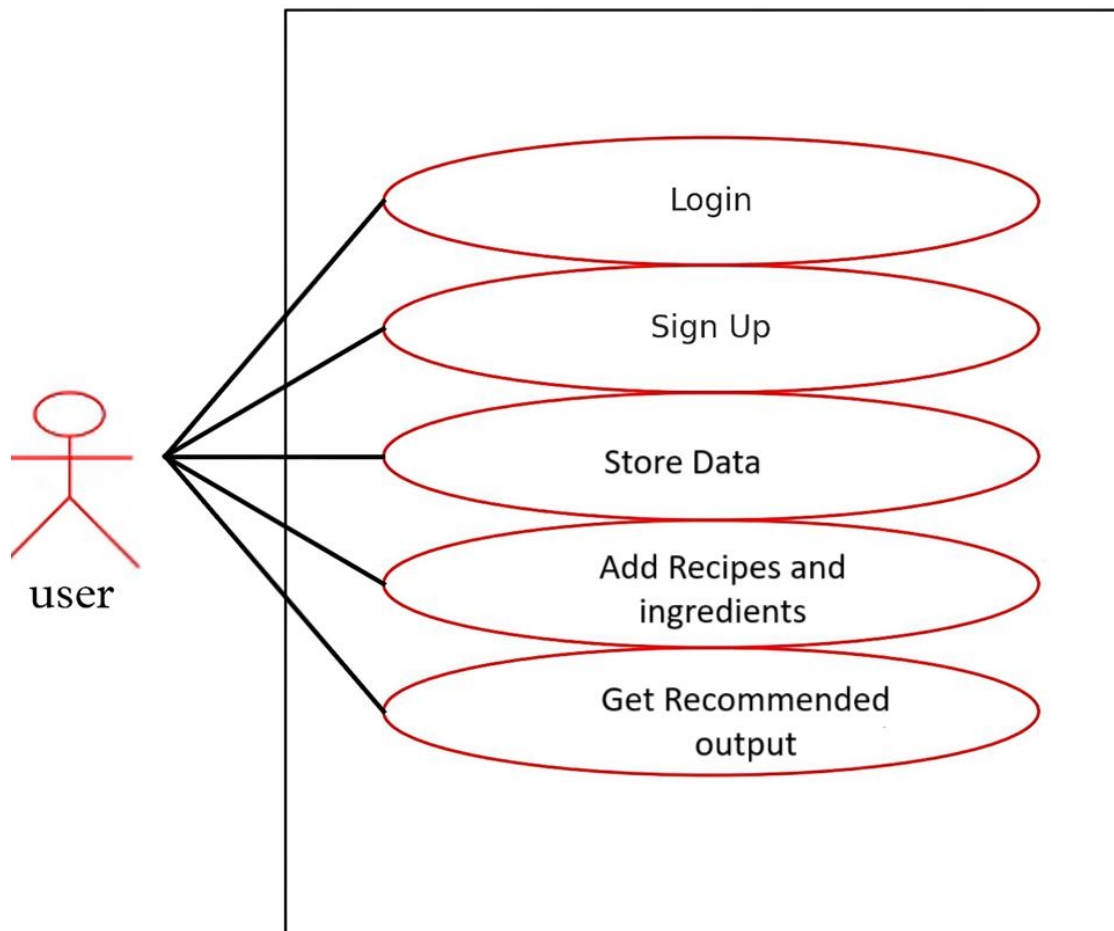
Following is a sample use case diagram representing the order management system. Hence, if we look into the diagram then we will find three use cases (Order, SpecialOrder, and NormalOrder) and one actor which is the customer. The SpecialOrder and NormalOrder use cases are extended from Order use case. Hence, they have extended relationship. Another important point is to identify the system boundary, which is shown in the picture. The actor Customer lies outside the system as it is an external user of the system.

USE CASE REGISTER



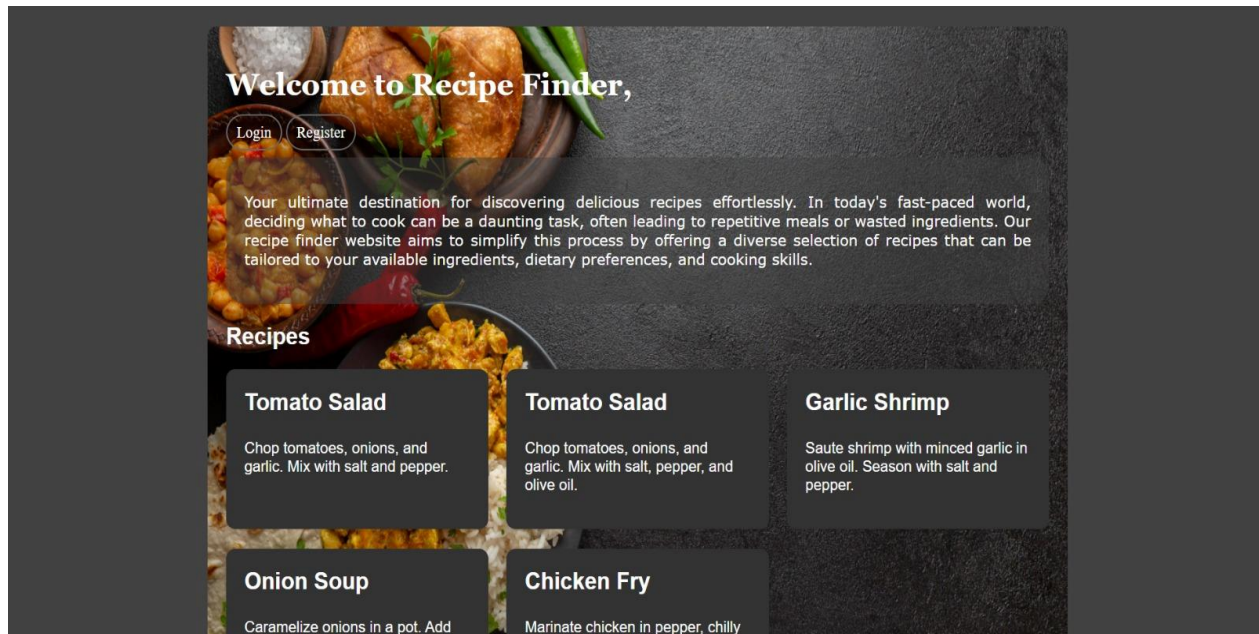
Use case is a methodology used in system analysis to identify, clarify and organize system requirements. The use case is made up of possible interactions between system and users in a particular environment and related to a particular goal. In the above diagram there are 6 classes: scan image, image acquisition, feature extraction, feature selection, classification, view result.

In the class scan image, the input image of the currency is captured by the user. The captured image is then acquired by the admin in the class image acquisition. In the class feature extraction and feature selection, the features of the image are extracted and relevant features are selected. In the class Classification the images are compared and the result is obtained as fake or genuine. In the view result class, the output is presented to the user in an audio format

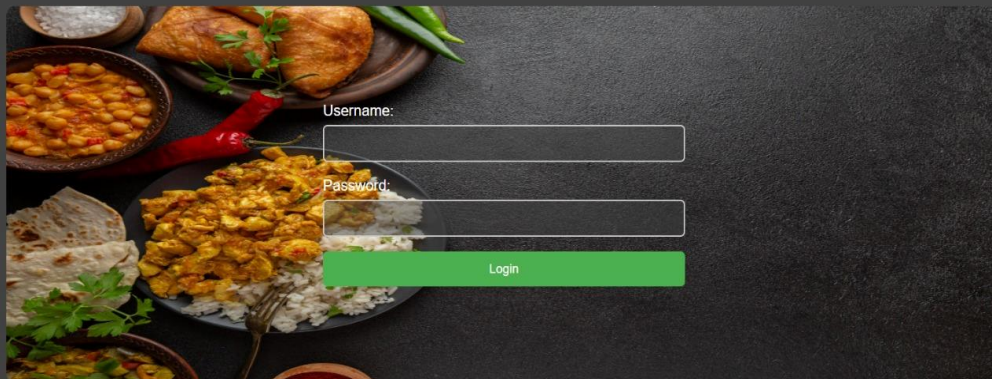


IMPLEMENTATION

5.1 SCREENSHOTS



Login to Recipe Finder

A login form is centered on a dark background featuring a collage of various Indian dishes, including a fried fish, a bowl of chickpeas, a plate of rice with a yellow curry, and some flatbread. The form consists of two input fields, a login button, and a copyright notice.

Username:

Password:

Login

© 2024 Recipe Finder

User Dashboard

Welcome, Group 10! [Logout](#)

Favorite Recipes

Onion Soup

Caramelize onions in a pot. Add broth, salt, and pepper. Simmer and serve.

[Remove from Favorites](#)

Find Recipes

Enter ingredients :

[Search](#)

Search Recipes

Enter recipe :

[Search](#)

Search Results

Tomato Salad

Chop tomatoes, onions, and garlic. Mix with salt and pepper.

▼ Add to Favorites

Tomato Salad

Chop tomatoes, onions, and garlic. Mix with salt, pepper, and olive oil.

▼ Add to Favorites

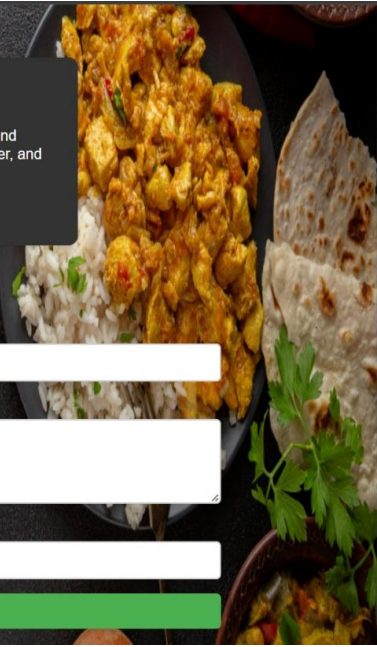
Add New Recipe

Recipe Name:

Instructions:

Ingredients (comma separated):

Add Recipe



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CHAPTER 6

Conclusion

The Recipe Finder project successfully developed a user-friendly platform where individuals can input their available ingredients and receive tailored recipe suggestions. This tool effectively reduces food waste, aids in meal planning, and encourages culinary exploration. Its responsive design ensures accessibility across various devices, making it a practical and valuable resource for everyday cooking.

REFERENCES

- <https://www.geeksforgeeks.org/-to-write-a-good-srs-for-your-project/>
- <https://github.com>
- <https://realfood.tesco.com/what-can-i-make-with.html>